

A P&ID is a complex representation of the various units found in a plant. It is used by people in a variety of crafts. The primary users of the document after plant startup are process technicians and instrument and electrical, mechanical, safety, and engineering personnel.

In order to read a P&ID, the technician needs an understanding of the equipment, instrumentation, and technology. In order to read a P&ID, the technician needs an understanding of the equipment, instrumentation, and technology. The next step in using a P&ID is to memorize your plant's process symbol list. This information can be found on the process legend. Process and instrument drawings have a variety of elements, including flow diagrams, equipment locations, elevation plans, electrical layouts, loop diagrams, title blocks and **legends**, and **foundation drawings**. The entire P&ID provides a three-dimensional look at the various operating units in a plant. The next step in using a P&ID is to memorize your plant's process symbol list.

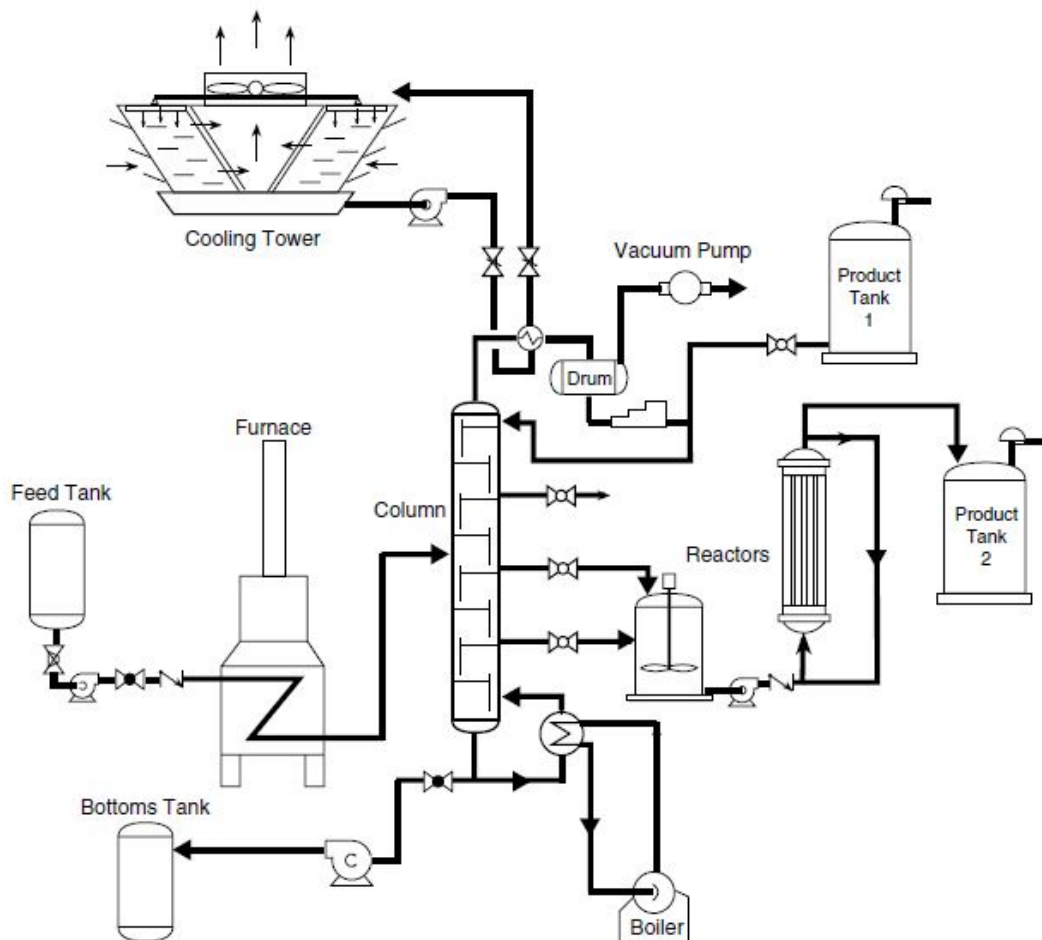
Key Terms

Electrical drawings—symbols and diagrams that depict an electrical process.

Elevation drawings—a graphical representation that shows the location of process equipment in relation to existing structures and ground level.

Equipment location drawings—show the exact floor plan for location of equipment in relation to the plan's physical boundaries.

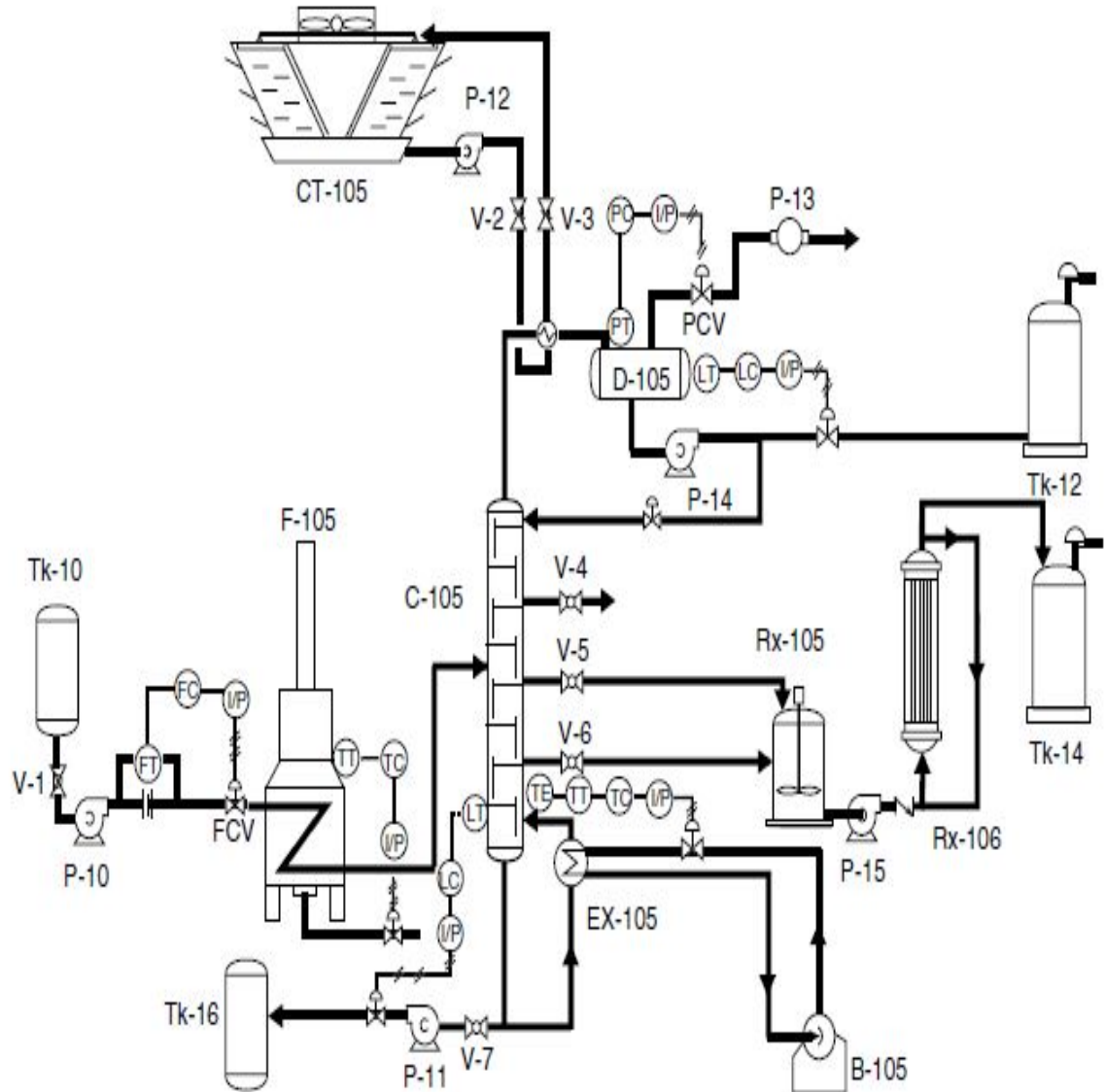
Flow diagram—a simplified sketch that uses symbols to identify instruments and vessels and to describe the primary flow path through a unit.



Foundation drawings—concrete, wire mesh, and steel specifications that identify width, depth, and thickness of footings, support beams, and foundation.

Legends—a document used to define symbols, abbreviations, prefixes, and specialized equipment. The process legend provides the information needed to interpret and read the P&ID. Process legends are found at the front of the P&ID. The legend includes information about piping, instrument and equipment symbols, abbreviations, unit name, drawing number, revision number, approvals, and company prefixes. Because symbol and diagram standardization is not complete, many companies use their own symbols in unit drawings. Unique and unusual equipment will also require a modified symbols file.

Process and instrument drawing (P&ID)—a complex diagram that uses process symbols to describe a process unit; also called piping and instrumentation drawing.

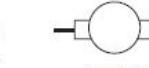
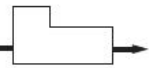
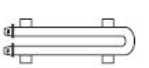
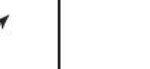


Types of Process Diagrams

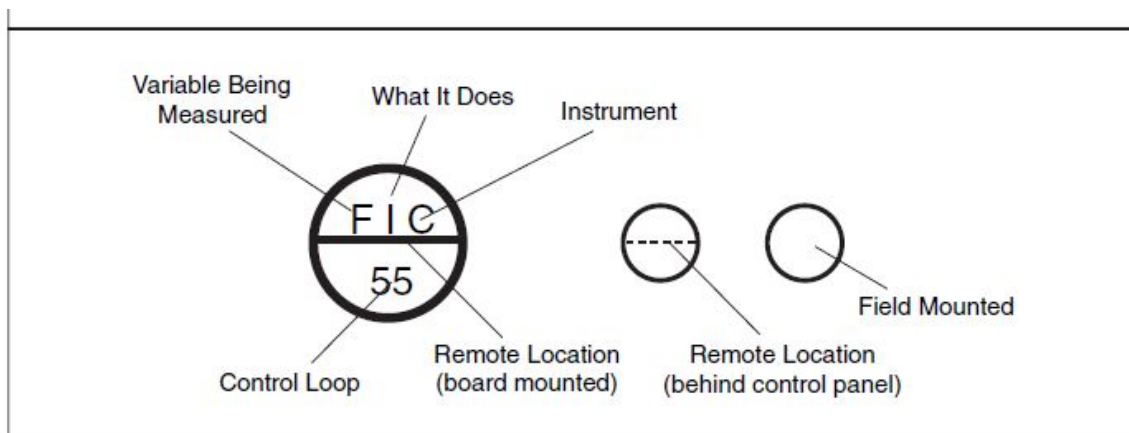
Process diagrams can be broken down into two major categories: process **flow diagrams** (PFDs) and **process and instrument drawings (P&IDs)**, sometimes called piping and instrumentation drawings. A flow diagram is a simple illustration that uses process symbols to describe the primary flow path through a unit. A process flow diagram provides a quick snapshot of the operating unit. Flow diagrams include all primary equipment and flows. A technician can use this document to trace the primary flow of chemicals through the unit. Secondary or minor flows are not included. Complex control loops and instrumentation are not included. The flow diagram is used for visitor information and new employee training. A process and instrument drawing is more complex. The P&ID includes a graphic representation of the equipment, piping, and instrumentation. Modern process control can be clearly inserted into the drawing to provide a process technician with a complete picture of electronic and instrument systems. Process operators can look at their process and see how the engineering department has automated the unit. Pressure, temperature, flow, and level control loops are all included on the unit P&ID.

Basic Instrument Symbols

Process technicians use P&IDs to identify all of the equipment, instruments, and piping found in their units.

VALVES  Gate Valve  Needle  Four-Way  Angle  Diaphragm  Manual Operated Valve  Gauge  Solenoid Valve CLOSED  Hydraulic  Back Pressure Regulator  Pneumatic Operated  Back Pressure Regulator  Globe Valve  Butterfly  Three-Way Valve  Ball  Check Valve  Plug  Bleeder Valves  Orifice  Pneumatic Operated Butterfly Valve  Motor  Rotameter  Relief PRV  Safety PSV											
COMPRESSORS  Reciprocating Compressor  Compressor & Silencers  Centrifugal Compressor  Rotary Compressor  Liquid Ring Compressor  Centrifugal Compressor (Turbine Driven)						PUMPS & TURBINE  Centrifugal Pumps  Vacuum Pump  Vertical  Gear Pump  Screw Pump  Turbine					
COMPRESSORS  Reciprocating Compressor  Compressor & Silencers  Centrifugal Compressor  Rotary Compressor  Liquid Ring Compressor  Centrifugal Compressor (Turbine Driven)						PUMPS & TURBINE  Centrifugal Pumps  Vacuum Pump  Vertical  Gear Pump  Screw Pump  Turbine					
HEAT EXCHANGERS  Hairpin Exchanger  U-Tube Heat Exchanger  Shell & Tube Heat Exchanger  Single Pass Heat Exchanger  Reboiler  Heater  Condenser						LINE SYMBOLS Future Equipment ----- Major Process _____ Minor Process _____ Pneumatic // // // // Hydraulic L L L Capillary Tubing X X X X X Electromagnetic Signal ~~~~~ Electric - - - - -					

 Temp Indicator	 Flow Indicator	 Transducer
 Temp Transmitter	 Flow Transmitter	 Pressure Indicating Controller
 Temp Recorder	 Flow Recorder	 Pressure Recording Controller
 Temp Controller	 Flow Controller	 Level Alarm
 Level Indicator	 Pressure Indicator	 Flow Element
 Level Transmitter	 Pressure Transmitter	 Temperature Element
 Level Recorder	 Pressure Recorder	 Level Gauge
 Level Controller	 Pressure Controller	 Analyzer Transmitter

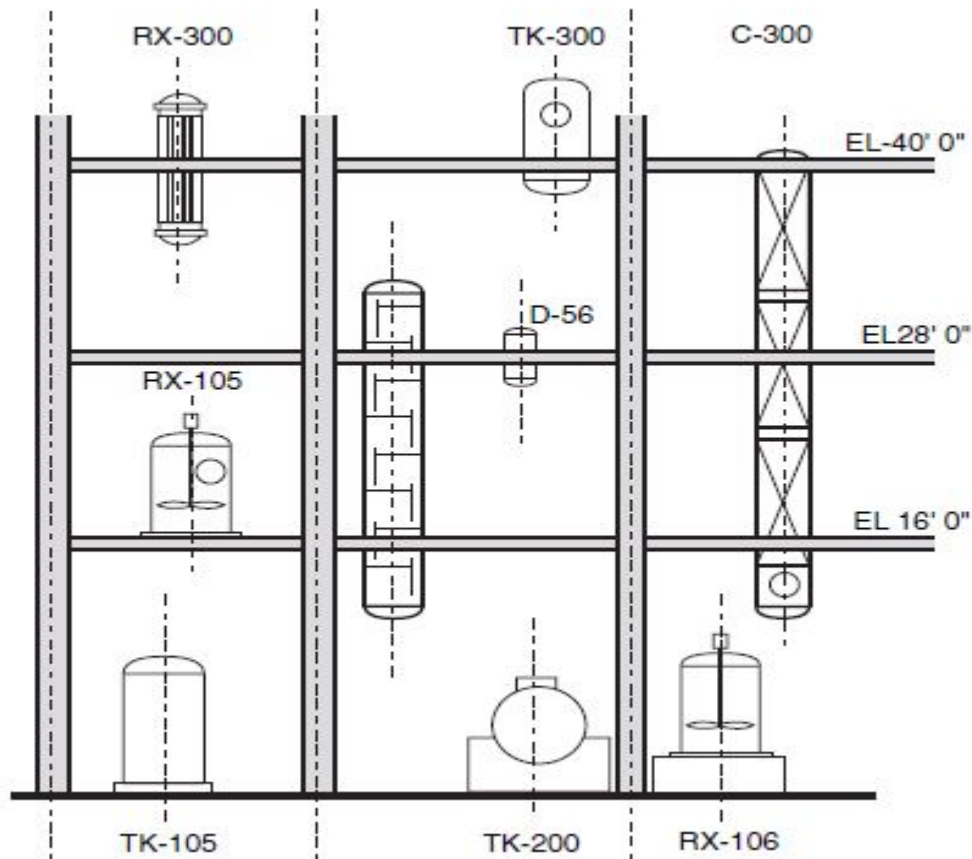


Foundation Drawing

The construction crew pouring the footers, beams, and foundation uses foundation drawings. Concrete and steel specifications are designed to support equipment, integrate underground piping, and provide support for exterior and interior walls. Process technicians do not typically use foundation drawings, but these drawings are useful when questions arise about piping that disappears under the ground and when new equipment is being added.

Elevation Drawing

Elevation drawings show the location of process equipment in relation to existing structures and ground level. In a multistory structure, the elevation drawing provides the technician with information about equipment location. This information is important for making rounds, checking equipment, developing checklists, catching samples, and performing startups and shutdowns.



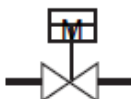
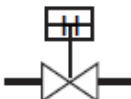
Loop Diagrams

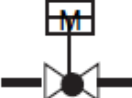
A loop diagram traces all instrument connections between the field instrument and the control room panel. This includes instrument air lines, wiring connections at field junction boxes, and control room panels and front connections.

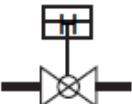
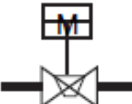
Review of Basic and Specialized Symbols

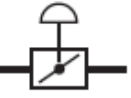
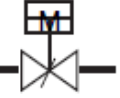
Piping and Valves

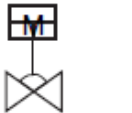
Each plant will have a standardized file for their piping symbols. Process technicians should carefully review the piping symbols for major and minor flows; electric, pneumatic, capillary, and hydraulic elements; and future equipment.

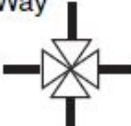
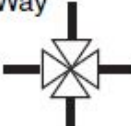
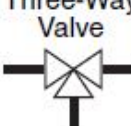
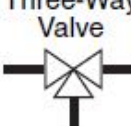
GATE VALVES					
					
Gate Valve	Manual Operated Valve	Pneumatic	Motor	Hydraulic	Bleeder Valves

GLOBE VALVES							
							
Globe Valve	Pneumatic	Motor	Hydraulic	Angle	Pneumatic	Motor	Needle

BALL VALVES			PLUG VALVES		
					
Ball	Ball	Motor or Hydraulic	Plug	Plug	Motor or Hydraulic

BUTTERFLY VALVES						
						
Butterfly	Butterfly	Butterfly	Motor or Hydraulic	Pneumatic Operated	Solenoid Valve CLOSED	

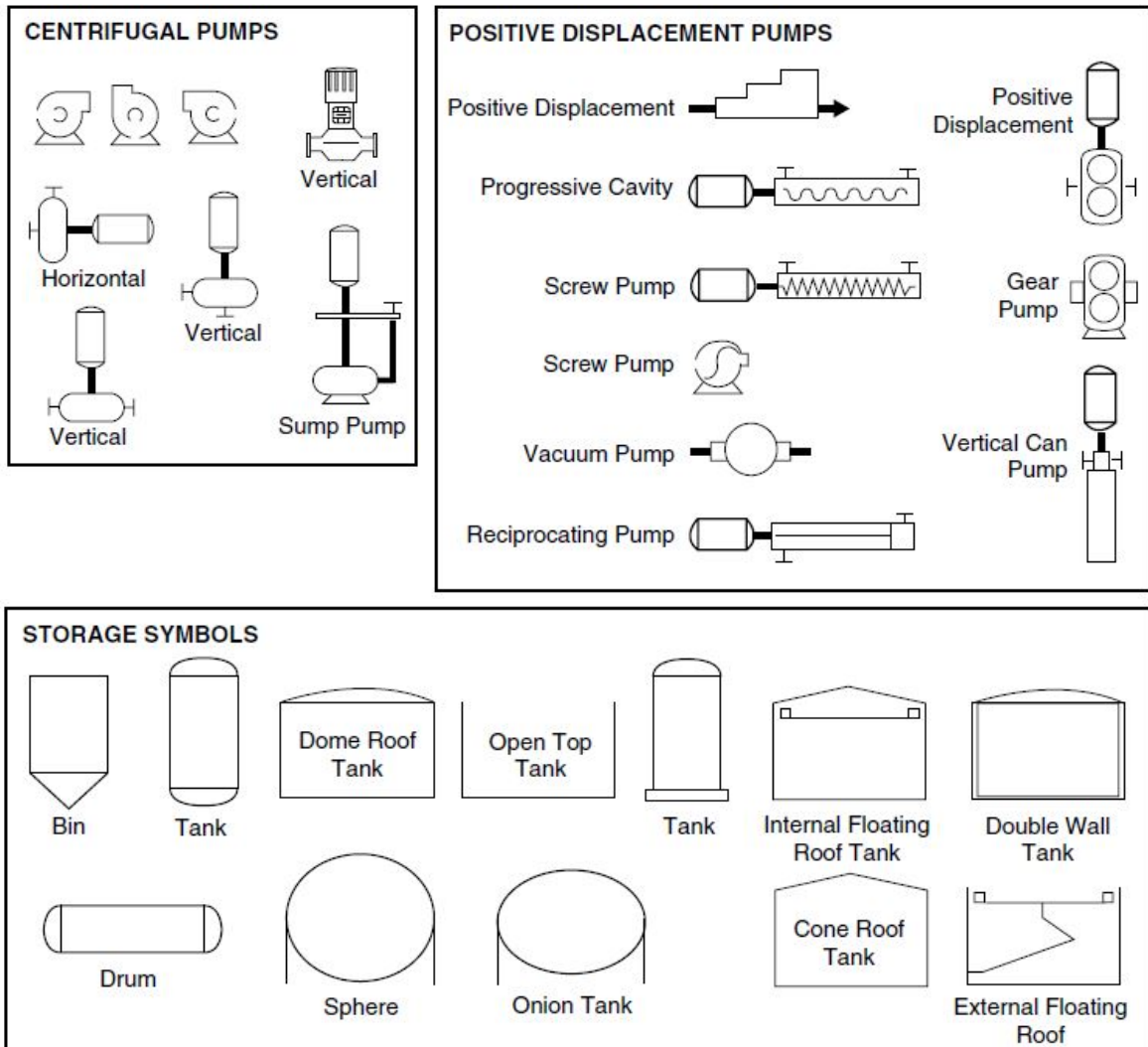
DIAPHRAGM VALVES			CHECK VALVES		
					
Diaphragm		Motor	Check Valve	Check Valve	Stop Check

SAFETY (Gases)		RELIEF (Liquids)		Four-Way		Three-Way Valve	
							
		Relief Valve	Relief Valve				

Knife Valve		Pinch Valve				
				Gauge	Rotameter	Orifice

Pumps and Tanks

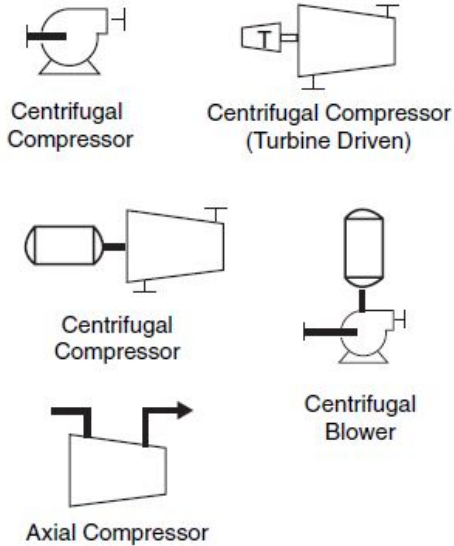
Pumps and tanks come in a variety of designs and shapes. Common pump and tank symbols are shown below.



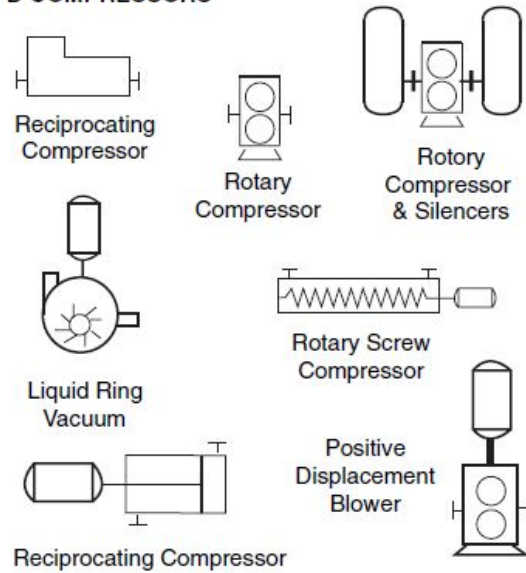
Compressors, Steam Turbines, and Motors

Because compressors and pumps share a common set of operating principles and are classified as dynamic or positive displacement, the symbols for compressors may closely resemble those for pumps.

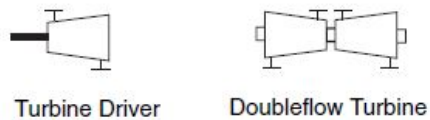
CENTRIFUGAL COMPRESSORS



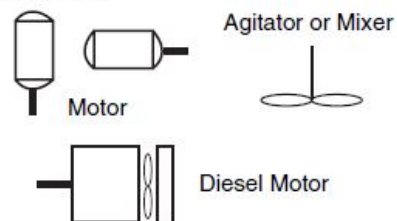
PD COMPRESSORS



STEAM TURBINE

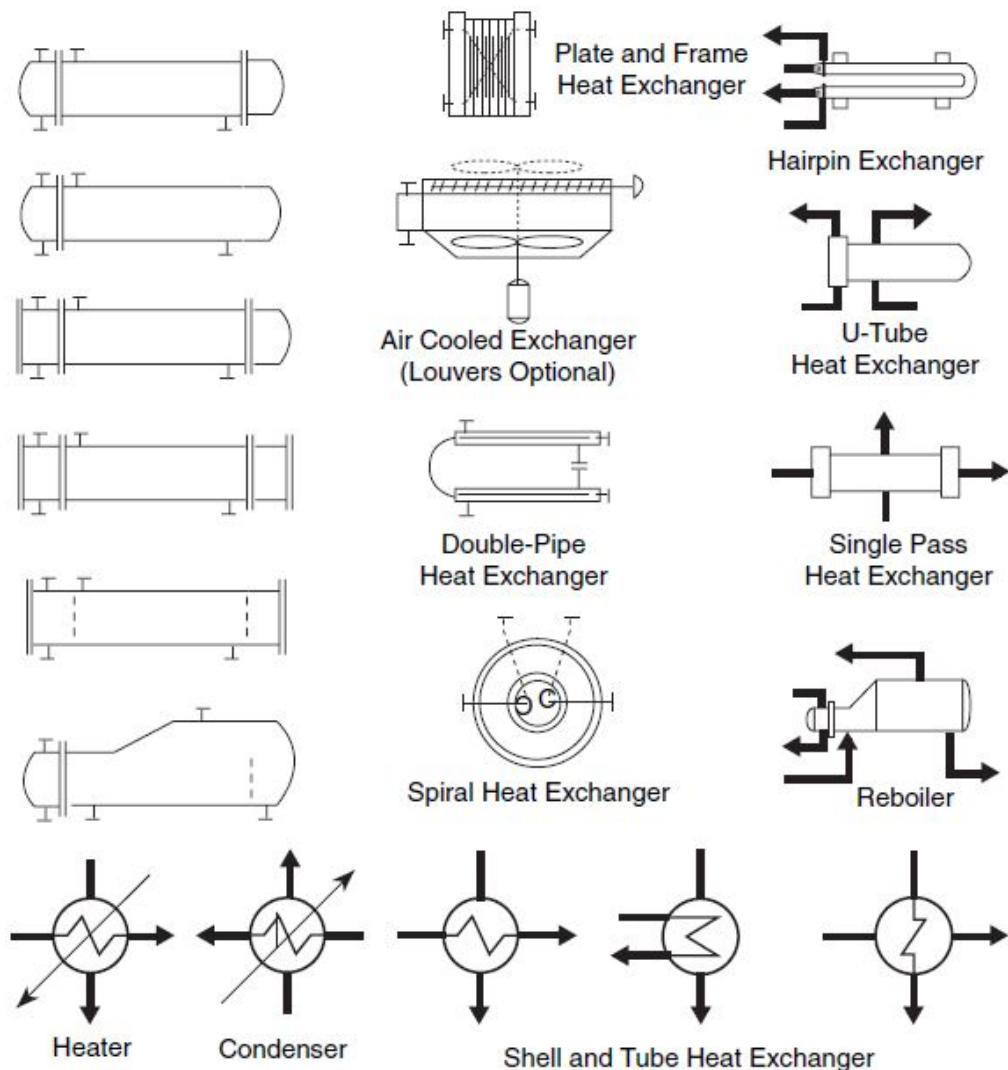


MOTORS



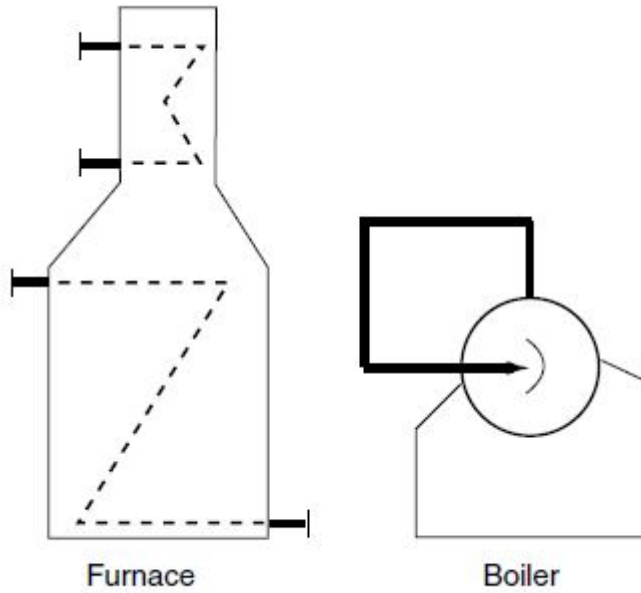
Heat Exchangers and Cooling Towers

Heat exchangers and cooling towers are two types of industrial equipment that share a unique relationship. A heat exchanger is a device used to transfer heat energy between two process flows. The cooling tower performs a similar function, but cooling towers and heat exchangers use different scientific principles to operate. Heat exchangers transfer heat energy through conductive and convective heat transfer, whereas cooling towers transfer heat energy to the outside air through the principle of evaporation. The symbol for a cooling tower is designed to resemble the actual device in the process unit. Cooled product flows out of the bottom of the tower and to the processing units. Hot water returns to a point located above the fill. The symbol will not show all of the various components of the cooling tower system, but it will provide a technician with a good foundation in cooling tower operation and enough information to clearly see the process.



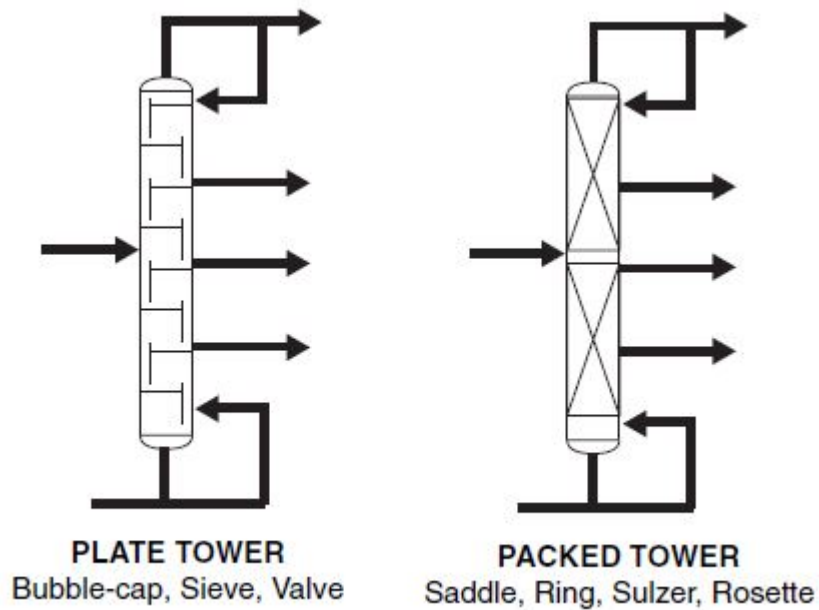
Furnaces and Boilers

The standard symbols file for furnaces and boilers is shown below. If a proprietary process includes several types of equipment not typically found on a standard symbol file, the designer will draw the device as it visually appears in the unit.



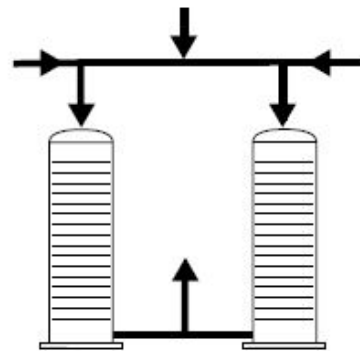
Distillation Columns

Distillation columns come in two basic designs, plate and packed. Flow arrangements vary from process to process. The symbols allow the technician to identify primary and secondary flow paths. Distillation is a process designed to separate the various components in a mixture by their boiling points. A distillation column is the central component of a much larger system. Plate distillation columns include sieve trays, valve trays, and bubble-cap trays. Packed columns are filled with packing material, rings, saddles, sulzer, and rosette.

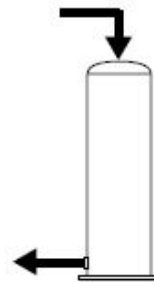


Reactors

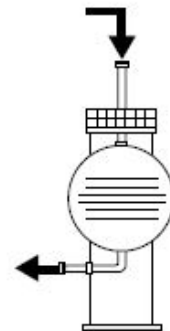
Reactors in the figure below are stationary vessels and can be classified as batch, semibatch, or continuous. A reactor is designed to allow chemicals to mix together under specific conditions to make chemical bonds, break chemical bonds, or make and break chemical bonds to form new products.



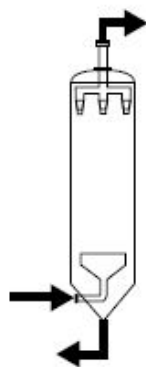
Hydrocracking



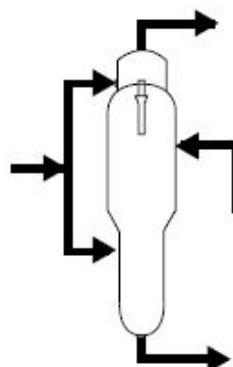
Hydrodesulfurization



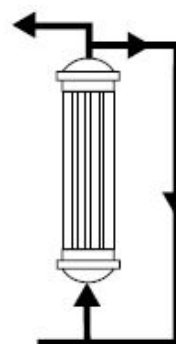
Reformer



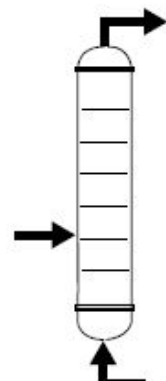
Fluid Catalytic
Cracking



Fluid Coking



Tubular
Reactor



Fluidized
Reactor

Sources of Information for Process Technicians

Information used by process technicians comes from a variety of sources.

Some of these sources are:

- Operating training manuals
- Process descriptions
- Process control manuals
- Equipment summaries
- Safety, health, and environment regulations
- Operating procedures
- Startup and shutdown procedures

Summary

Process flow diagrams (PFDs) and process and instrument drawings (P&IDs) are used to outline or explain the complex flows, equipment, instrumentation, electronics, elevations, and foundations that exist in a process unit. A PFD is a simple flow diagram that describes the primary flow path through a unit. A P&ID is a complex representation of the various units found in a plant. Standardized symbols and diagrams have been developed for most pieces of industrial equipment, process flows, and instrumentation.